

House Price Prediction Report

TABLE OF CONTENT

1. Problem Statement	2
2. Need for Solving This Problem	2
3. Data Description Details	3
4. EDA and Business Implications	
4.1. Univariate Analysis	(4 – 5)
4.2. Bivariate Analysis	(6-10)
4.3 Multi-variate Analysis	(10-11)
5. Data Cleaning and Pre- Processing	
5.1. Missing Value Treatment	12
5.2. Need for variable transformation (if any)	13
5.3. Variables removed or added and why (if any)	14
6. Model Building	
6.1. Model Selection	14
6.2. Performance Improvement	15
7. Model Validation	16
8. Final Interpretation & Recommendation.....	(17 -18)

FIGURES

Fig 1. Data Description	3
Fig 2. Dataset Information	4
Fig 3. Histogram of year build chart	5
Fig 4. Histogram of Bathroom chart	5
Fig 5. Furnished vs. Price Scatter chart	6
Fig 6. Coast vs. Price Scatter chart	6
Fig 7. Bedrooms vs. Price chart	6

Fig 8. Bathrooms vs. Price chart	7
Fig 9. Living Measure vs. Price chart	7
Fig 10. Floors vs. Price chart	7
Fig 11. Sight vs. Price chart	8
Fig 12. Condition vs. Price chart	8
Fig 13. Condition vs. Price chart	8
Fig 14. Ceiling Measure vs. Price chart	9
Fig 15. Basement Size vs. Price chart	9
Fig 16. Total_area vs Price chart	10
Fig 17. House_sold_date vs. Price chart	10
Fig 18. Heatmap Chart	11
Fig 19. Check Null value	12
Fig 20. Check Outlier	13
Fig 21. Post treatment of Outlier	13
Fig 22. Model Details	14

1 - Introduction - > Determining the right price for a house is a crucial challenge for homeowners, buyers, and real estate businesses. A house's value is influenced by multiple factors beyond just location and square footage, including the number of bedrooms, amenities, neighbourhood quality, market trends, and more. Pricing a property too high may result in a prolonged selling period, while pricing it too low could lead to financial loss.

Traditionally, homeowners and real estate agents rely on comparable properties in the same neighbourhood to estimate a house's price. However, this approach can be subjective and may not account for all variables affecting property values. With advancements in data science and machine learning, building a predictive model for house pricing can provide a more accurate, data-driven estimation.

2 - Need for Solving This Problem

- **Improved Pricing Decisions:** A well-trained model helps sellers and buyers make informed decisions by estimating the fair market price.
- **Reducing Uncertainty:** Provides a more systematic approach than relying solely on neighbourhood comparisons.

- **Market Insights:** Helps investors and real estate agencies identify price trends and potential investment opportunities.
- **Efficiency in Transactions:** Accurate predictions streamline the buying and selling process by reducing negotiation time and pricing discrepancies.

Main Objective - By leveraging data-driven techniques, we can develop a robust regression model to predict house prices, improving transparency and accuracy in the real estate market.

3 - Data Description:

No	Variable	Feature description
1	cid	A notation for a house
2	dayhours	Date house was sold
3	price	Price is prediction target
4	room_bed	Number of Bedrooms/House
5	room_bath	Number of bathrooms/bedrooms
6	living_measure	square footage of the home
7	lot_measure	quare footage of the lot
8	ceil	Total floors (levels) in house
9	coast	House which has a view to a waterfront
10	sight	Has been viewed
11	condition	How good the condition is (Overall)
12	quality	Grade given to the housing unit, based on grading system
13	ceil_measure	Square footage of house apart from basement
14	basement_measure	Square footage of the basement
15	yr_built	Built Year
16	yr_renovated	Year when house was renovated
17	zipcode	Zip
18	lat	Latitude coordinate
19	long	Longitude coordinate

20	living_measure15	Living room area in 2015(implies-- some renovations) This might or might not have affected the lot size area
21	lot_measure15	Lot size area in 2015(implies-- some renovations)
22	furnished	Based on the quality of room
23	total_area	Measure of both living and lot

Features Details -

We have below feature in our dataset.

- 4 features are of integer type
- 12 features are of float type
- 7 feature is of object type

```
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RangeIndex: 21613 entries, 0 to 21612
Data columns (total 23 columns):
#   Column                Non-Null Count  Dtype
---  -
0   cid                   21613 non-null  int64
1   dayhours              21613 non-null  object
2   price                 21613 non-null  int64
3   room_bed              21505 non-null  float64
4   room_bath             21505 non-null  float64
5   living_measure        21596 non-null  float64
6   lot_measure           21571 non-null  float64
7   ceil                  21571 non-null  object
8   coast                 21612 non-null  object
9   sight                 21556 non-null  float64
10  condition             21556 non-null  object
11  quality               21612 non-null  float64
12  ceil_measure          21612 non-null  float64
13  basement              21612 non-null  float64
14  yr_built              21612 non-null  object
15  yr_renovated          21613 non-null  int64
16  zipcode               21613 non-null  int64
17  lat                   21613 non-null  float64
18  long                  21613 non-null  object
19  living_measure15      21447 non-null  float64
20  lot_measure15         21584 non-null  float64
21  furnished             21584 non-null  float64
22  total_area            21584 non-null  object
dtypes: float64(12), int64(4), object(7)
memory usage: 3.8+ MB
```

4. EDA and Business Implications -

4.1 - Univariate Analysis -

Basement: Highly skewed; most properties have no basement, with a few large outliers.

Bathrooms: Majority have 1-2 bathrooms; properties with more than 2 are outliers.

Bedrooms: Most homes have 3-4 bedrooms, with rare extreme cases up to 30.

Ceil Measure: Right skewed, with most values between 1000-3000 sq. ft.; some extreme outliers.

Condition: Most properties rate 3 or 4; very few have extreme low/high scores.

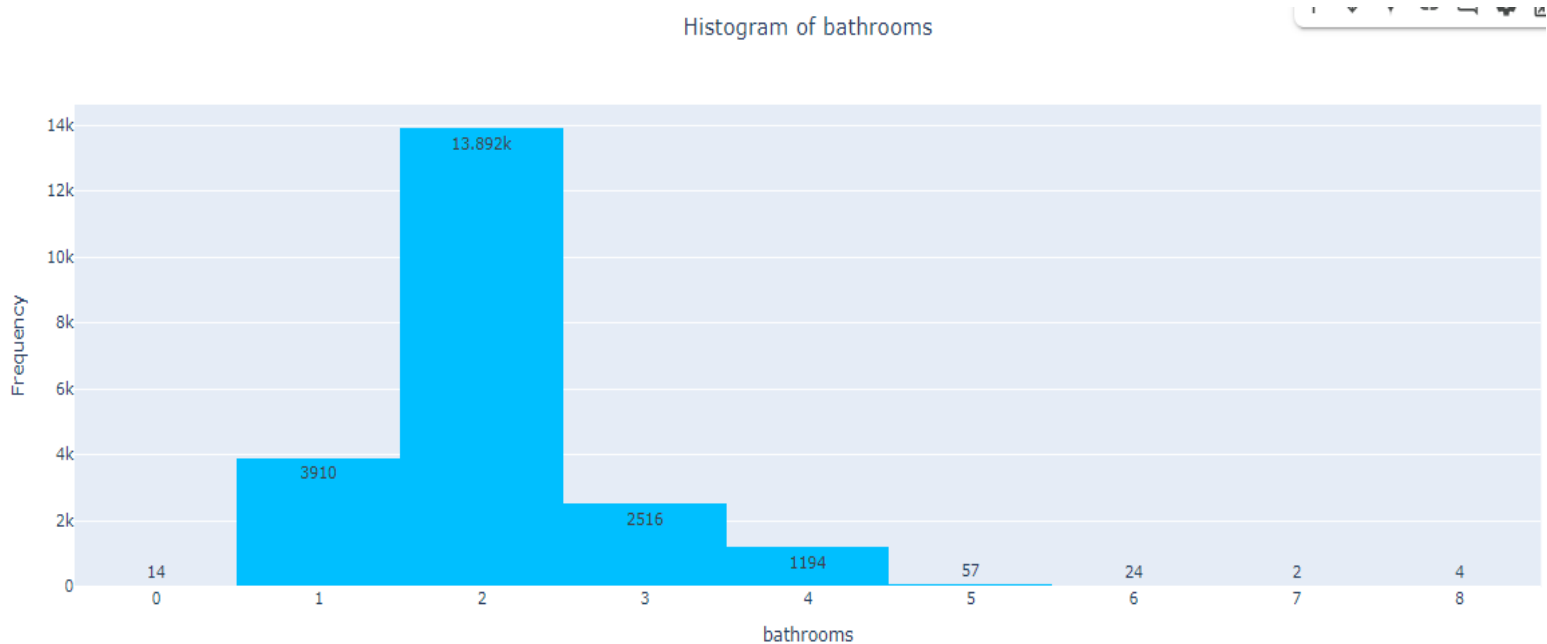
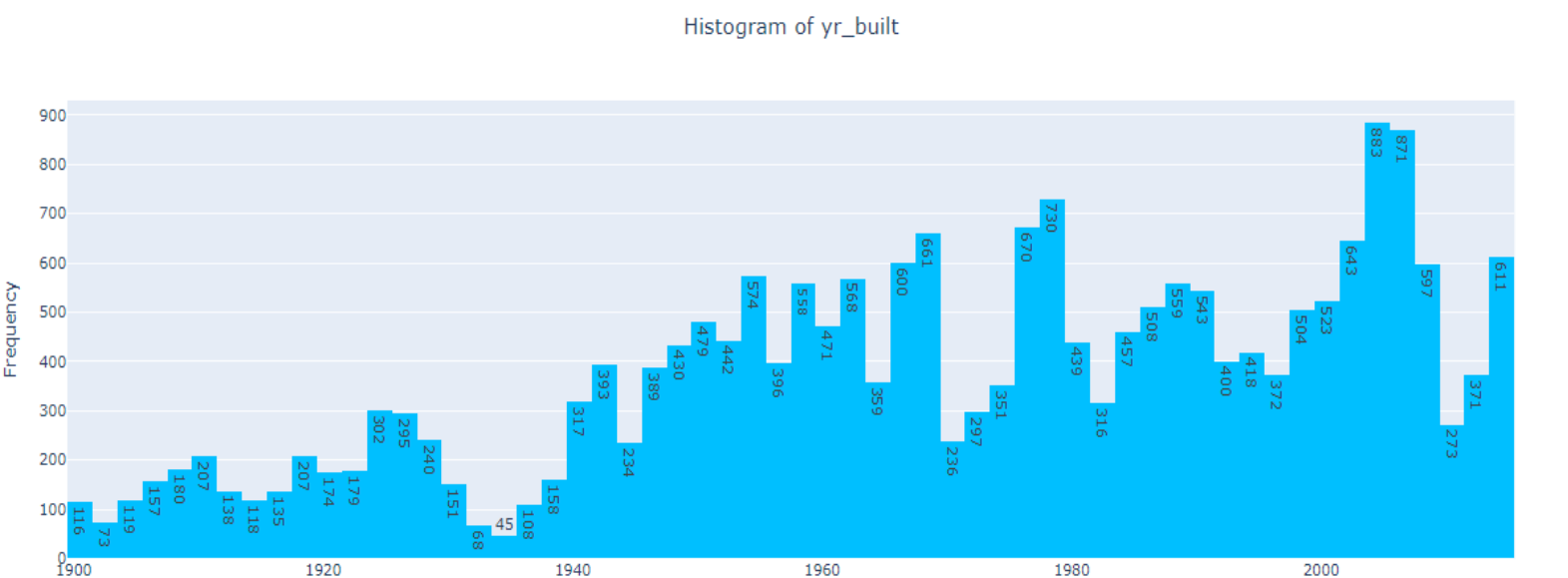
Floors: Mostly 1-2 floors, with very few reaching 3, no major outliers.

Living Measure: Right skewed, with most homes between 1000-2500 sq. ft.; some exceed 4000.

Price: Right skewed; most homes below \$1M, with luxury properties exceeding \$3M as outliers.

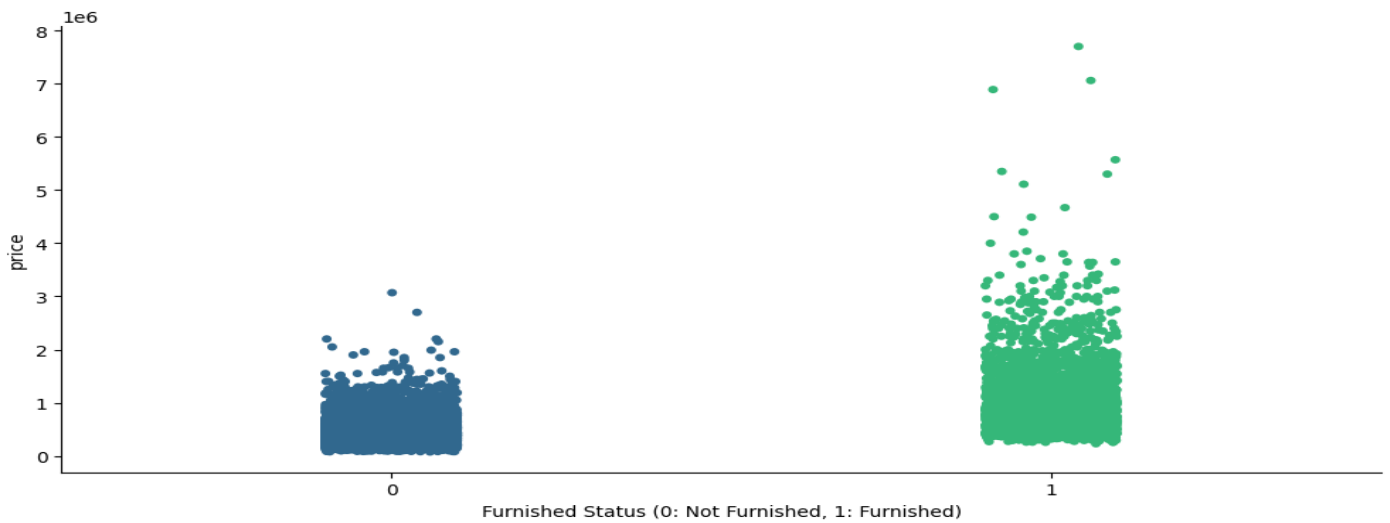
Total Area: Right skewed; most properties are smaller, with some extremely large outliers.

Year Built: Most homes built post-1950, peaking in the 1970s and early 2000s



4.2 - Bivariate Analysis:

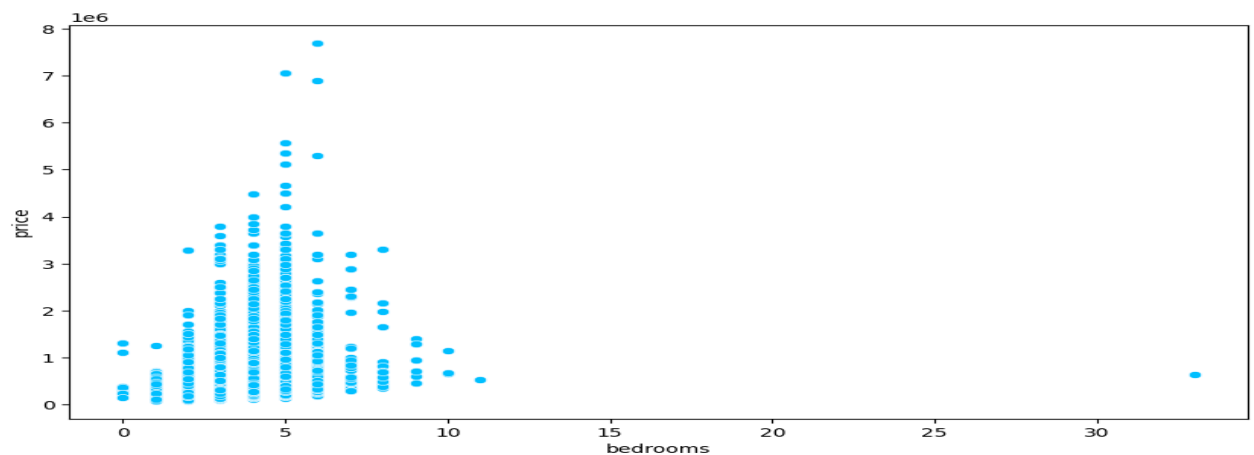
1. Furnished vs. Price: Furnished houses tend to be more expensive.



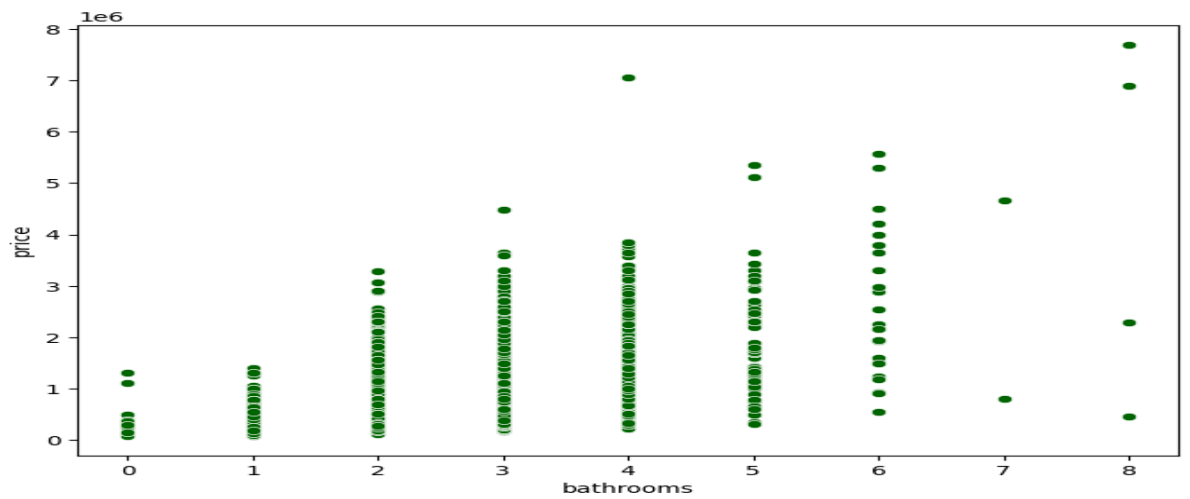
2. Coast vs. Price: Waterfront properties have significantly higher prices.



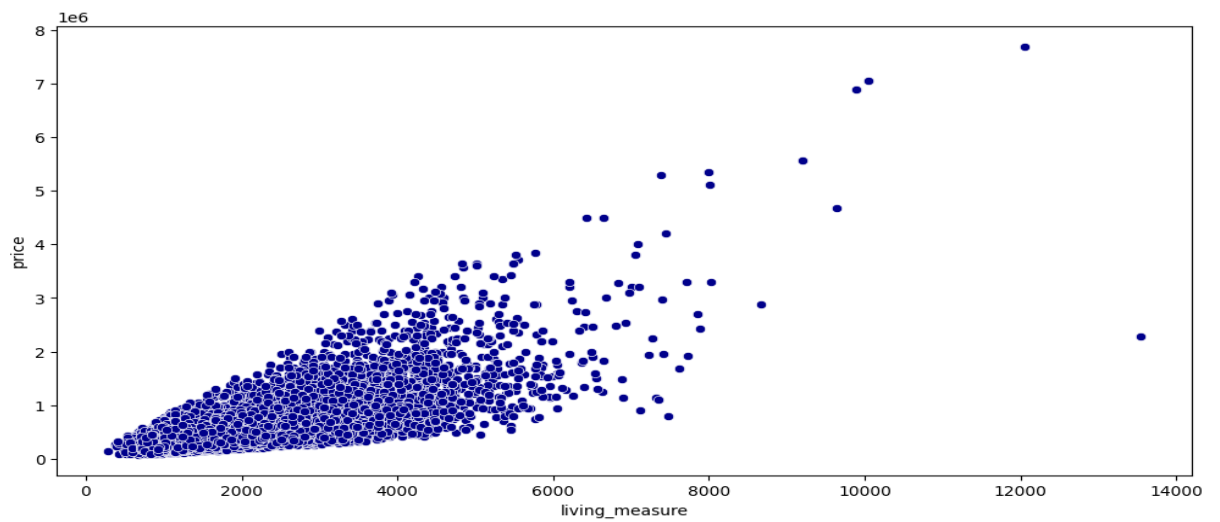
3. Bedrooms vs. Price: Prices vary widely for 3-6 bedrooms, but additional bedrooms don't always increase price consistently.



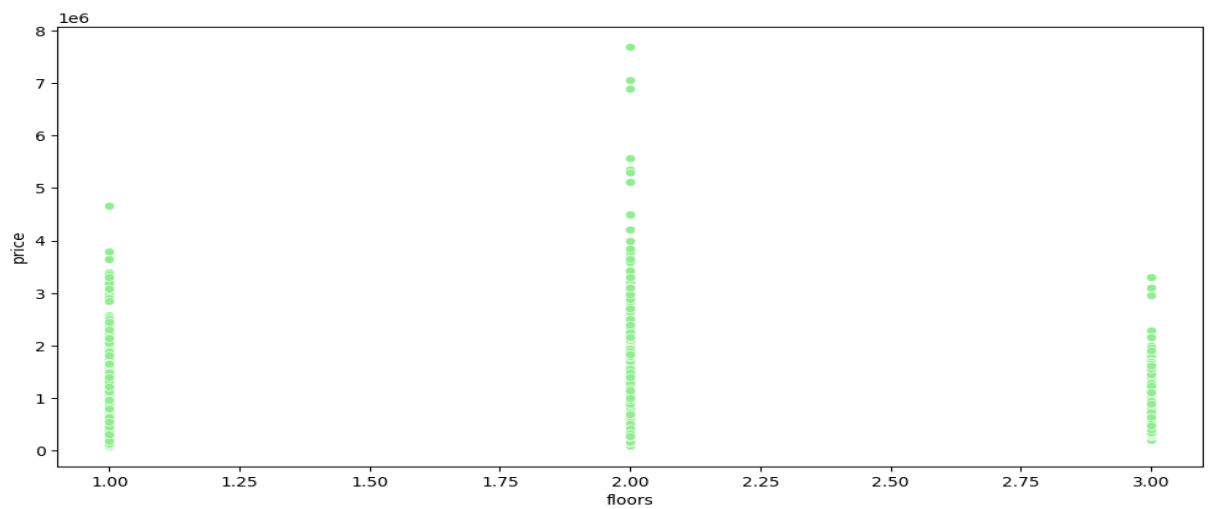
4. Bathrooms vs. Price: More bathrooms generally increase price, but variability exists.



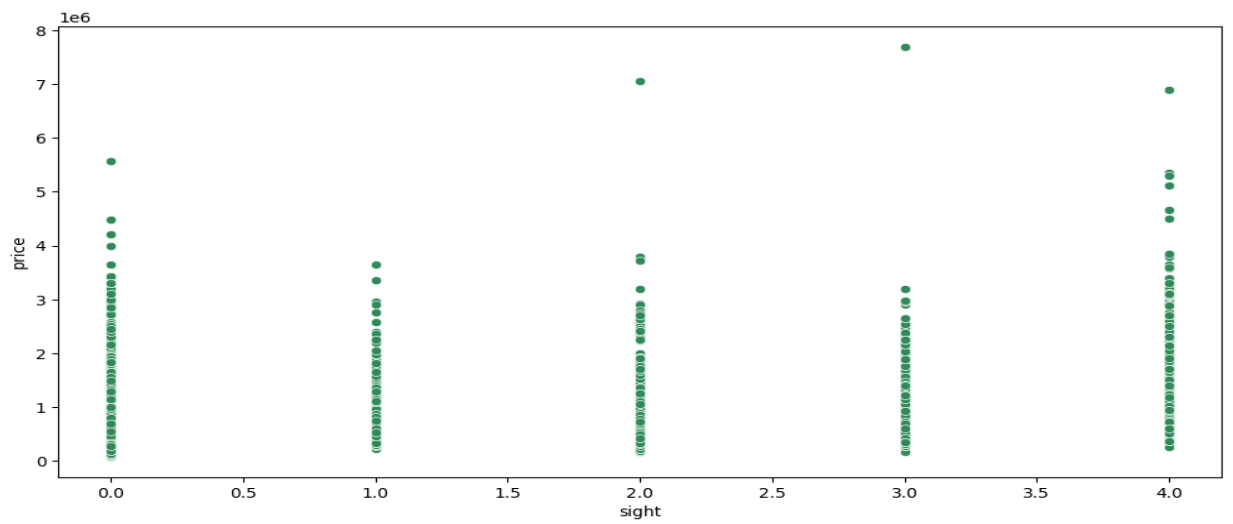
5. Living Measure vs. Price: Larger living areas correlate with higher prices, though outliers exist.



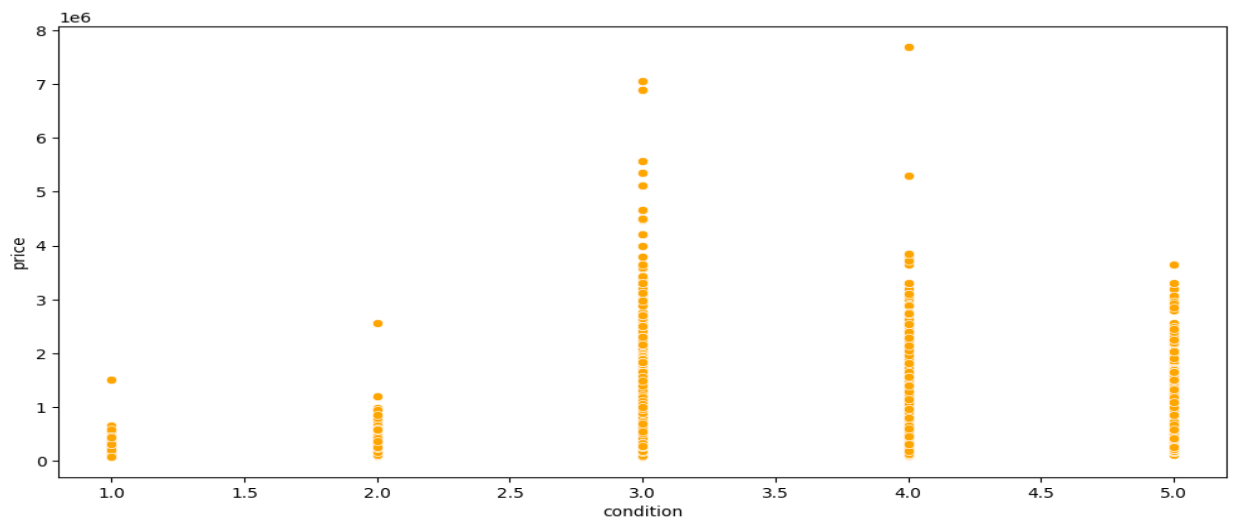
6. Floors vs. Price: Two-floor homes have the highest average prices.



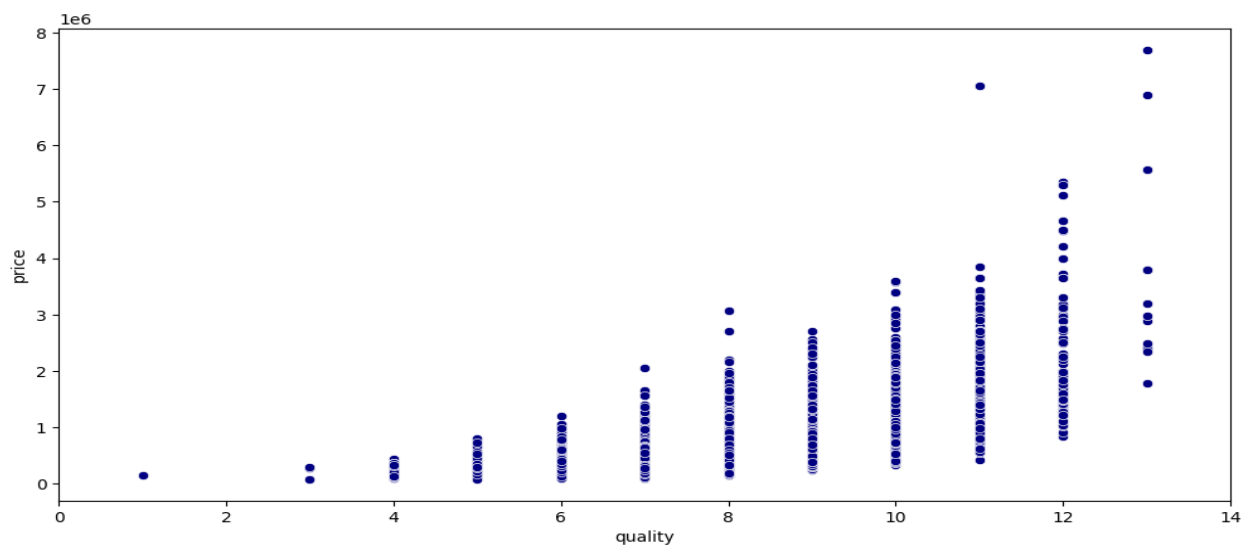
7. Sight vs. Price: Better views correlate with higher prices, such properties are rare.



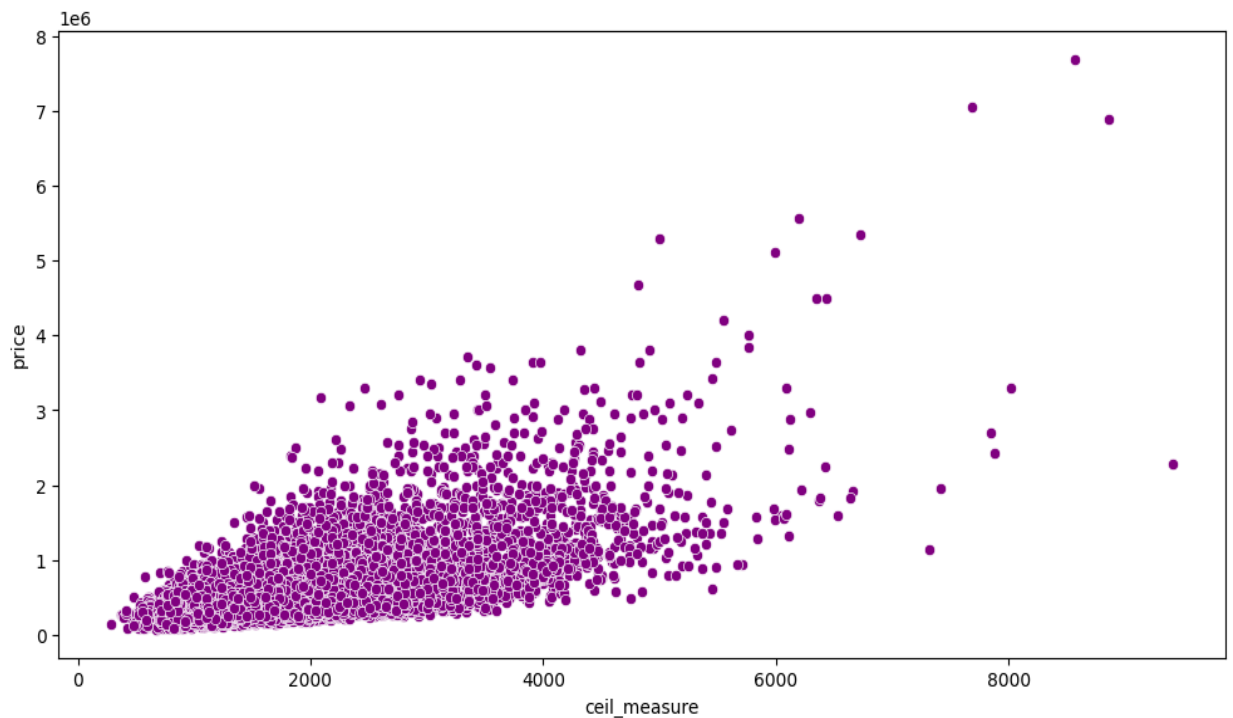
8. Condition vs. Price: Higher condition ratings generally lead to higher prices.



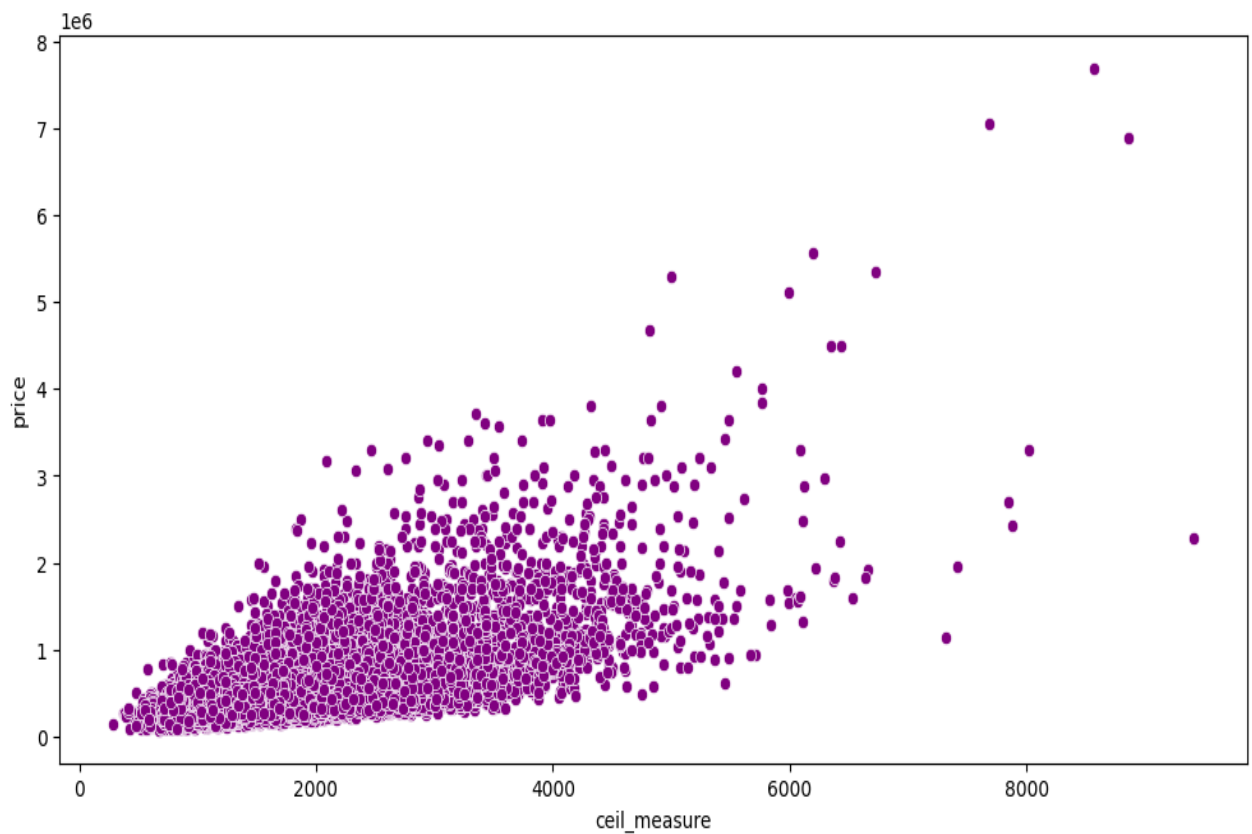
9. Quality vs. Price: Higher quality homes (10+) command significantly higher price.



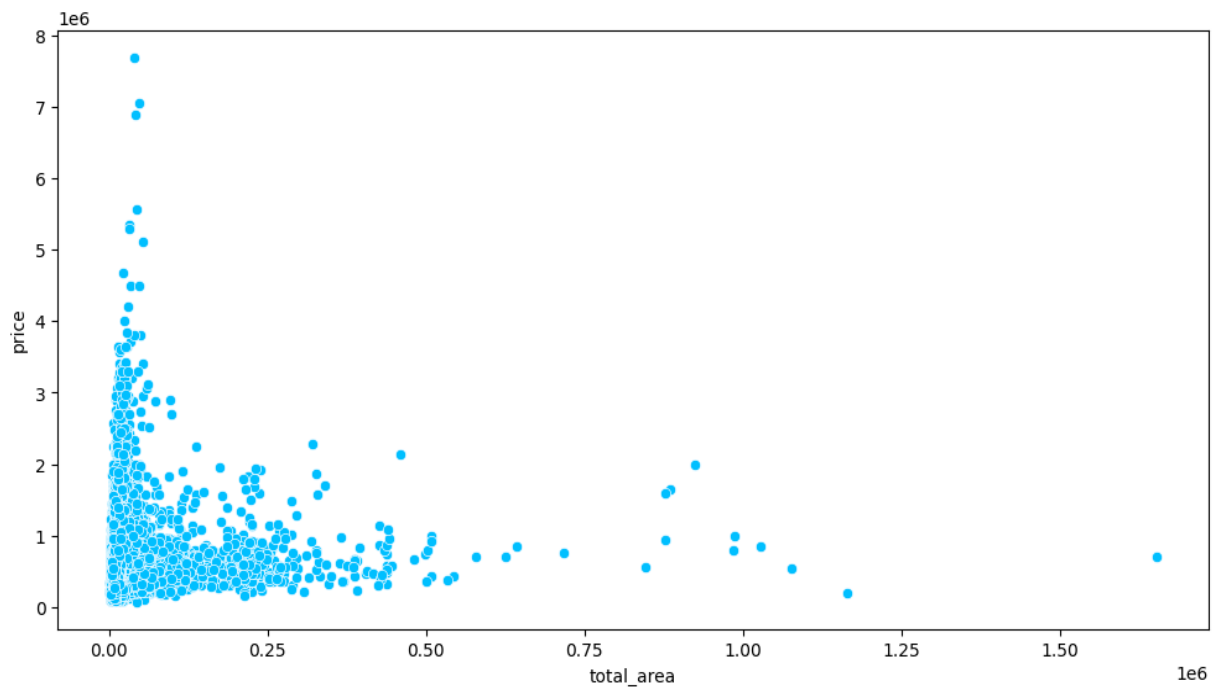
10. Ceiling Measure vs. Price: Taller ceilings are linked to higher prices, with some extreme outliers.



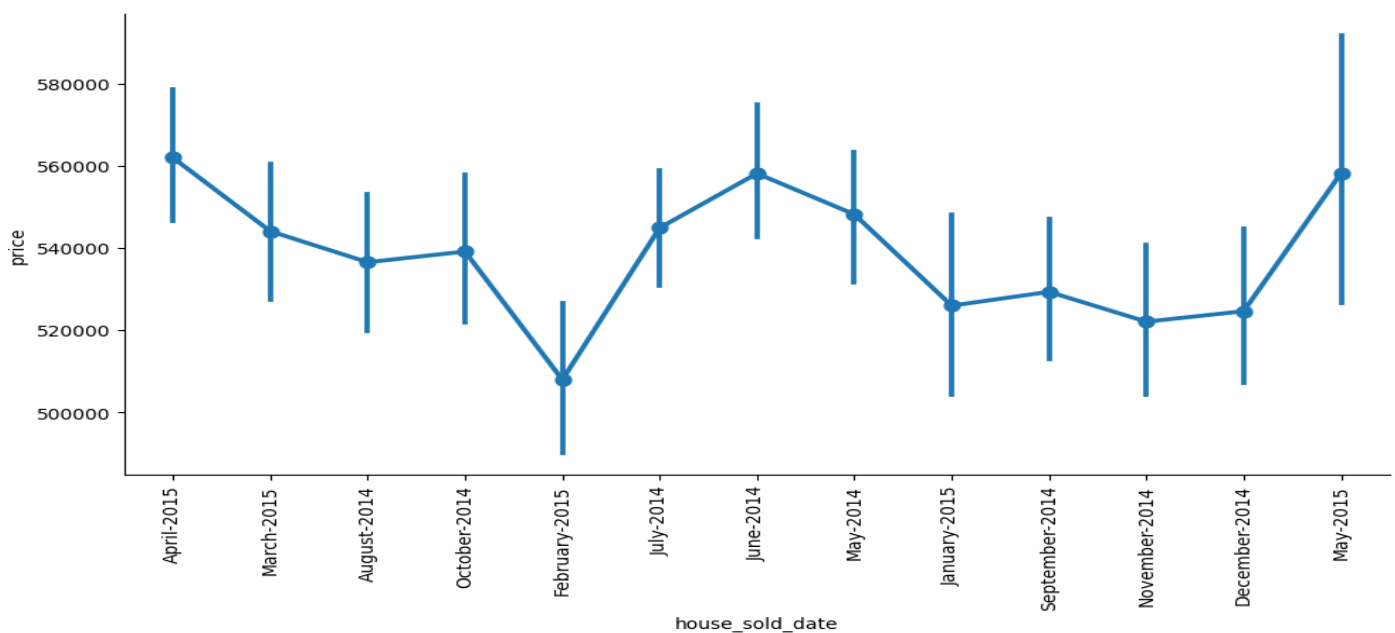
11. Basement Size vs. Price: Larger basements correlate with higher prices; many homes have no basement.



12. Total_area vs Price: There's a positive correlation between total_area and price, although the spread of prices suggests other factors may also significantly influence price.



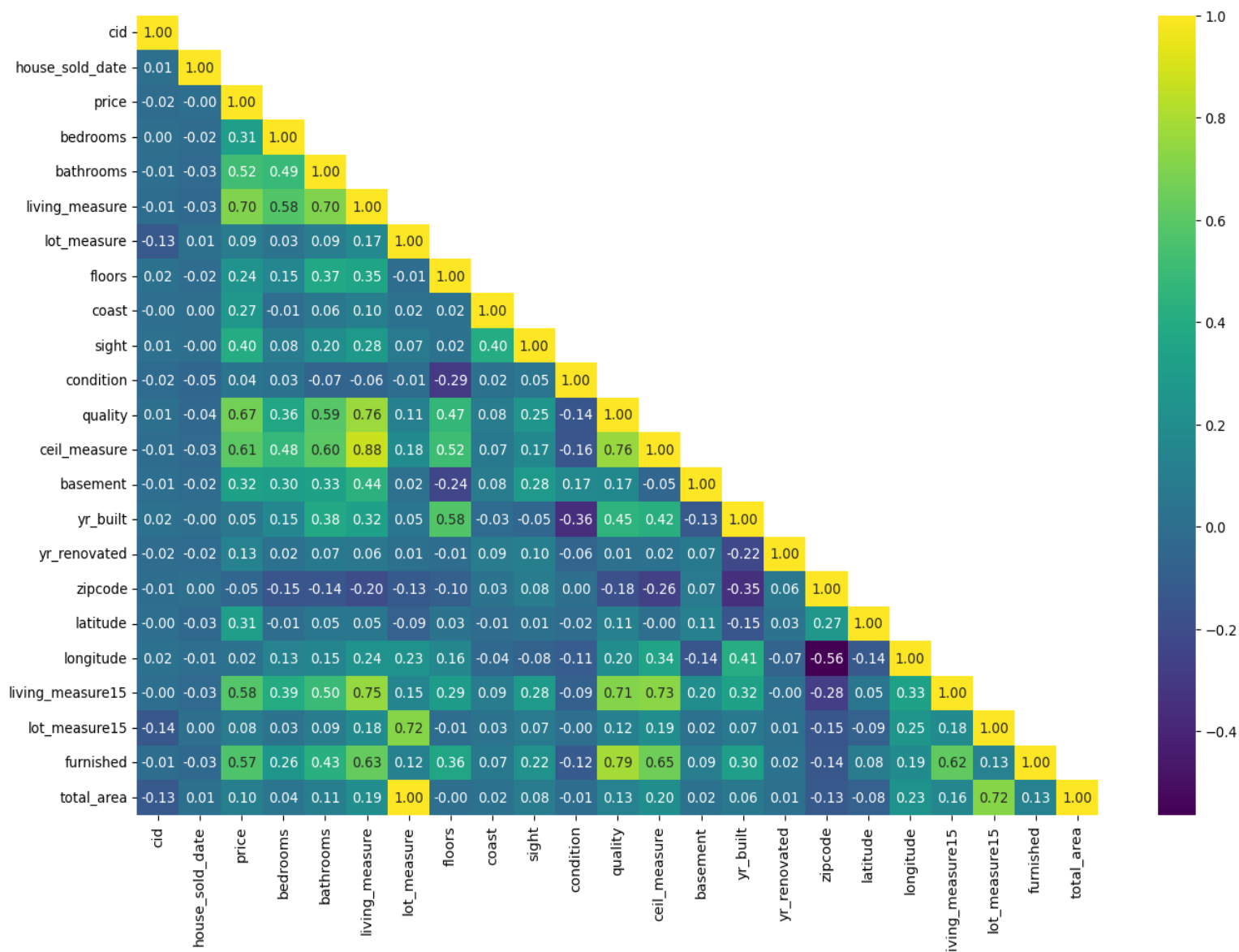
13. House_sold_date vs. Price: Prices fluctuated seasonally, peaking in April 2015 and June 2014.



4.3 - Multivariate Analysis -

We used heatmaps to analyse variable interactions affecting house prices.

- Price Correlation: Strongly linked to quality (0.67), living measure (0.70), and ceiling measure (0.64). Moderate links with bathrooms (0.52), floors (0.37), sight (0.40), and coast (0.27).
- Living & Total Area: Living measure correlates with ceiling measure (0.88) and quality (0.76), while total area is influenced by living measure (0.72) and lot measure (0.62).
- Bathrooms vs. Bedrooms: Bathrooms (0.52) impact price more than bedrooms (0.31).
- Year Built & Renovated: Year built (0.18) has a slight impact, while renovations (0.02) show minimal influence.
- Other Observations: Condition (-0.06) shows a weak negative impact, and location features have low direct correlations.



Heatmap Chart

5 - Data Cleaning and Pre-processing – Data Cleaning and Pre-processing is the process of handling missing values, removing duplicates, correcting inconsistencies, and transforming data into a suitable format for analysis or modelling. It ensures data quality, improves accuracy, and enhances model performance.

5.1 - Approach used for identifying and treating missing values and outlier

treatment (and why) -

Handling Missing Values: Missing values were treated using imputation techniques like mean/median/Model replacement and predictive imputation.

We have also identified there are no duplicate values in dataset.

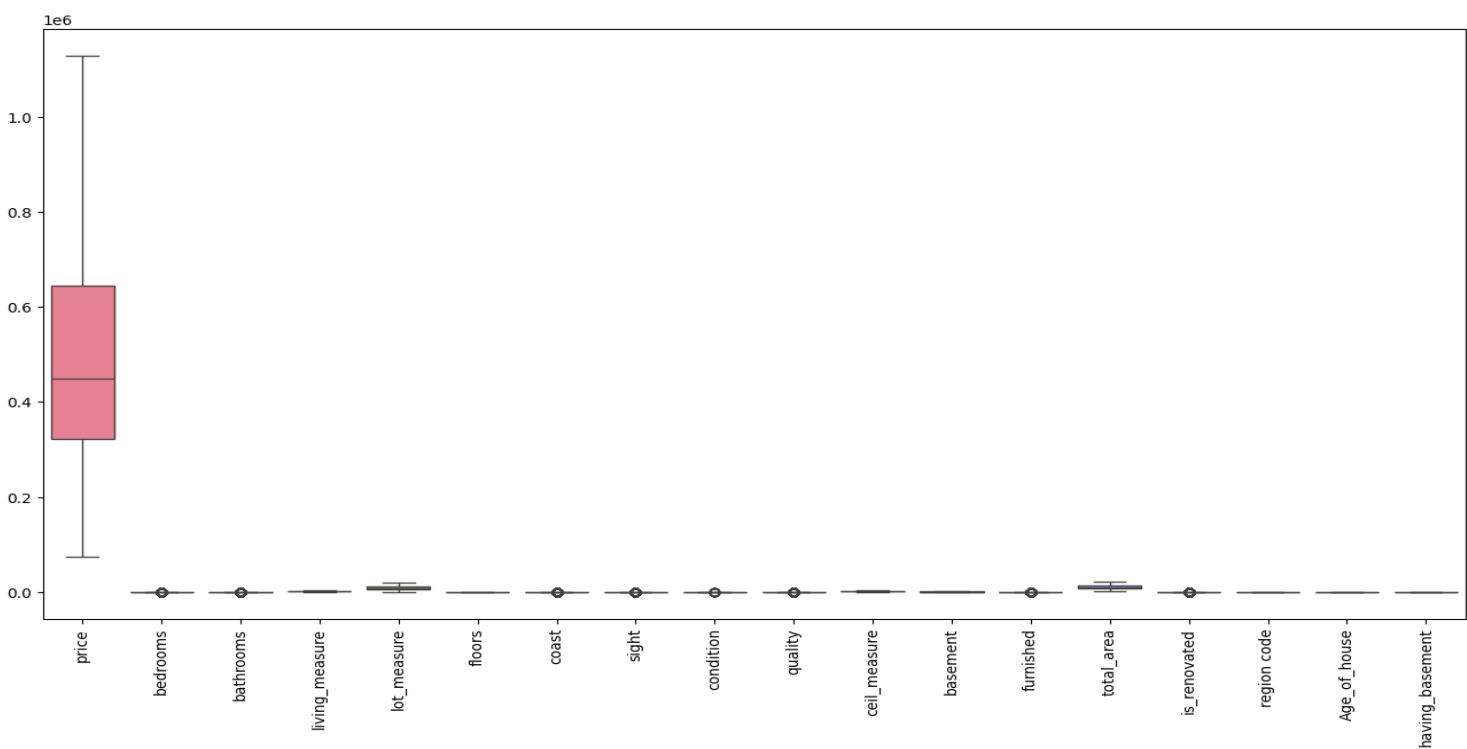
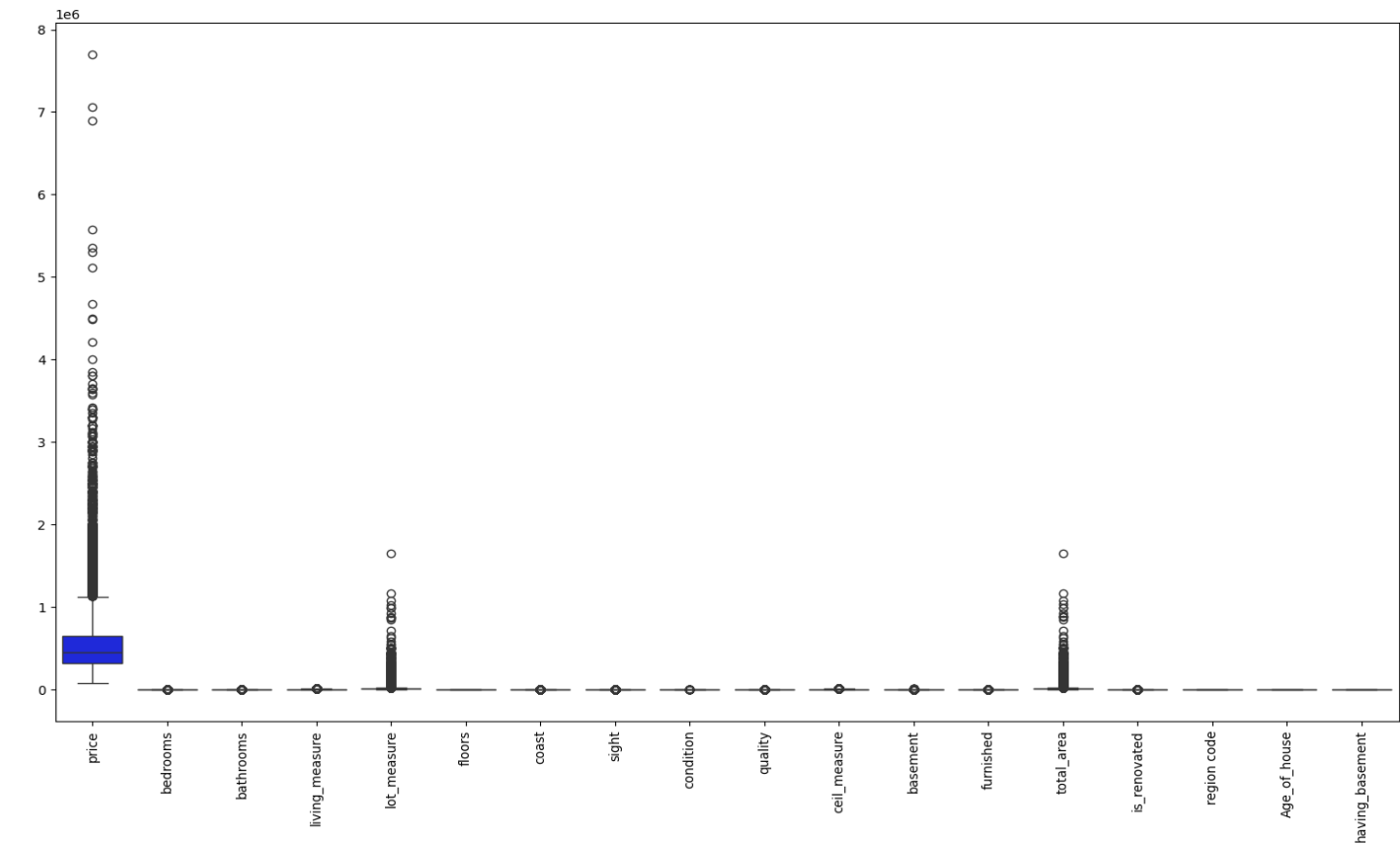
living_measure15	166
room_bed	108
room_bath	108
sight	57
lot_measure	42
furnished	29
lot_measure15	29
living_measure	17
quality	1
ceil_measure	1
basement	1
yr_renovated	0
long	0
lat	0
zipcode	0
cid	0
yr_built	0
dayhours	0
condition	0
coast	0
...	0

Outlier Treatment: Outliers were detected using box plots and we treated the outlier using IQR (Interquartile Range) technique.

Post Treatment - Based on the box plot you provided, here are some key observations:

1. Price Outliers: There are several outliers in the price column, with some houses priced significantly higher than the majority.
2. Living Measure Outliers: The living_measure feature also shows a considerable number of outliers, indicating some homes have much larger living areas compared to the rest.
3. Lot Measure Outliers: The lot_measure column exhibits a few extreme outliers, which means there are properties with exceptionally large lot sizes.
4. Total Area Outliers: Similar to the living_measure and lot_measure, the total_area feature has outliers indicating homes with unusually large total areas.
5. Other Features: For most of the other features like bedrooms, bathrooms, floors, coast, sight, condition, quality, etc., the data points are more concentrated around the median, with fewer or no significant outliers.

Overall, the outliers in price, living_measure, lot_measure, and total_area might require further investigation or potential handling depending on the analysis objective.



5.2 - Need for variable transformation (if any)

We have used standard scaler to transform them to have a mean of 0 and a standard deviation of 1.

5.3 - Variables removed or added and why (if any)

a - Remove Columns in Dataset – Irrelevant or highly correlated and no useful features were removed to improve model efficiency.

1. cid
2. longitude
3. latitude
4. house_sold_date
5. zipcode
6. yr_built
7. yr_renovated
8. living_measure15
9. lot_measure15
10. region code
11. basement

b - Add New Columns in Dataset – We have created some new columns in dataset for further Analysis.

- 1 - is_renovated
- 2 - region code
- 3 - years old
- 4 - is_basement

6 - Model Building -

6.1 - Model Selection: We have used various models which is mentioned below.

1. Linear Regression
2. Ridge Regression
3. Lasso Regression
4. Decision Tree Regression
5. Random Forest Regression
6. Support Vector Regression

7. Gradient Boosting,
8. XGBoost
9. AdaBoost

Model Performance Explanation

- **Gradient Boosting** performs the best with an **R^2 score of 0.746650**, indicating strong predictive performance. It balances bias and variance well, making it a reliable choice.
- **Random Forest Regression** also performs well (**$R^2 = 0.741256$**) but has a slightly lower test score compared to Gradient Boosting. While it captures patterns effectively, it may slightly overfit due to its high training score.
- **Ridge Regression and Linear Regression** have similar performance (**$R^2 \approx 0.7138$**), showing they are decent models but not as powerful as ensemble methods. Ridge Regression adds regularization to reduce overfitting.
- **Lasso Regression** also performs close to Ridge and Linear Regression, slightly improving feature selection but not significantly outperforming other models.
- **Decision Tree Regression** shows signs of **overfitting**, with a much higher training score (0.660082) compared to its test score (0.685557). While it learns the training data well, it struggles with generalization.
- **Support Vector Regression (SVR)** has a **low R^2 score of 0.635689**, meaning it doesn't perform as well as other models and struggles to capture the patterns in the dataset.
- **AdaBoost** has the lowest test score (**$R^2 = 0.618450$**), indicating it may not be the best choice for this problem due to its relatively high bias.

Final Conclusion: Gradient Boosting is the best-performing model due to its highest test score, strong generalization ability, and low error metrics.

	Model_Name	Training_Score	Test_Score	MSE	RMSE	MAE	R-squared
6	Gradient_Boost	0.733679	0.746650	1.577017e+10	125579.331163	92415.707605	0.746650
4	Random_Forest_Regression	0.725254	0.741256	1.610594e+10	126909.183233	92591.598515	0.741256
1	Ridge_Regression	0.702296	0.713846	1.781210e+10	133461.989347	101712.419205	0.713846
0	Linear_Regression	0.702290	0.713832	1.781298e+10	133465.286612	101714.834050	0.713832
2	Lasso Regression	0.702348	0.713787	1.781575e+10	133475.637924	101725.536591	0.713787
3	Decision_Tree	0.660082	0.685557	1.957298e+10	139903.471495	102888.837240	0.685557
5	Support_Vector_Regression	0.593838	0.635689	2.267709e+10	150589.131003	115696.592379	0.635689
7	AdaBoost	0.606959	0.618450	2.375017e+10	154110.901085	123404.765532	0.618450

6.2 - Performance Improvement: To enhance model performance, we implemented several advanced techniques, including feature engineering, hyperparameter tuning, ensemble learning, and cross-validation:

1. **Feature Engineering:** We applied Standard Scaler to normalize numerical features, ensuring that all variables are on the same scale. This step prevents models from being biased toward features with larger magnitudes and improves the efficiency of gradient-based optimization algorithms.
2. **Hyperparameter Tuning:** To optimize model performance, we utilized GridSearchCV, which systematically explores different combinations of hyperparameters to identify the best configuration. This approach ensures that our models operate at their highest potential by selecting optimal values for parameters such as learning rate, tree depth, and regularization factors.
3. **Ensemble Learning: Bagging & Boosting:** We leveraged ensemble techniques to enhance predictive accuracy and reduce variance:
Bagging (Bootstrap Aggregating): Used in models like Random Forest, where multiple decision trees are trained on different subsets of the data, and their predictions are averaged to improve robustness and reduce overfitting.
Boosting: Implemented with Gradient Boosting, XGBoost, and AdaBoost, where models are trained sequentially, with each iteration correcting the errors of the previous one. This method significantly improves performance by focusing on difficult-to-predict instances.
4. **Cross-Validation & Resampling:** To ensure reliable model evaluation and prevent overfitting, we applied K-Fold Cross-Validation. This technique splits the dataset into multiple folds, training the model on different subsets and evaluating it on the remaining data. By averaging the performance across all folds, we obtained a more generalized and robust model.

7 - Model Validation

- The model was validated using multiple metrics:
 - ✚ Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) to evaluate prediction accuracy.
 - ✚ R-squared (R^2) to assess how well the model explains the variance in house prices.
 - ✚ Cross-validation to ensure model generalization on unseen data.
 - ✚ We have also used K-Fold Cross-Validation.

6. Final Interpretation & Recommendation -

1. Data-Driven Pricing Strategy

- Leverage the Gradient Boosting model to set competitive prices based on property attributes such as quality, living area, ceiling height, and bathroom count, which show the highest correlation with price.
- Adjust pricing based on regional trends, ensuring properties in high-demand areas are priced accordingly.
- Regularly update the model with external factors such as market demand trends, interest rates, and economic indicators to maintain accuracy.

2. Addressing Data Imbalance for Fair Market Insights

- Waterfront properties (0.7%) and premium homes are underrepresented—adjust pricing strategies to reflect their exclusivity and high demand.
- Right-skewed distributions (price, lot size, living area) indicate a market catering primarily to middle-income buyers. Ensure high-end properties are properly marketed to affluent clientele.
- **Imbalance solutions:**
 - Use oversampling (SMOTE) or under sampling to balance dataset representation.
 - Apply weighted metrics in model training to give rare categories (e.g., waterfront homes) fair representation.
 - Cluster properties into affordable, mid-range, and luxury segments for more tailored insights.

3. Market Segmentation & Business Insights Using Clustering

- Affordable Homes: Smaller properties for first-time buyers and budget-conscious investors.
- Mid-Range Homes: Suburban properties with moderate pricing and strong demand.
- Luxury Homes: High-value properties featuring waterfront views, large living spaces, and premium amenities.

Implementation:

- Targeted Marketing: Highlight value-for-money homes and exclusive properties for niche buyers.

- Regional Focus: Direct inventory and promotional efforts toward high-demand clusters.
- Price Optimization: Identify undervalued properties and adjust pricing accordingly.

4. Maximizing Property Value with Key Feature Enhancements

- Invest in renovations that impact price the most:
 - Quality upgrades (0.67 correlation with price)
 - Living area expansion (0.70 correlation with price)
 - Bathroom additions (0.52 correlation with price, more than bedrooms)
- Avoid unnecessary renovations: Features like year built (0.18 correlation) and condition (-0.06 correlation) have minimal impact.

5. Inventory Management & Seasonal Strategy

- Peak demand months (e.g., April and June) should be leveraged for maximum sales impact.
- Ultra-luxury homes with extreme features (large lot size, high pricing) should be marketed as exclusive investments.
- Properties built around the year 2000 are in high demand—use this insight to guide acquisition and development strategies.

Conclusion

By implementing these data-driven recommendations, real estate agencies can optimize pricing, improve market segmentation, and maximize profitability. The use of Gradient Boosting, combined with resampling, clustering, and strategic renovations, ensures a robust and future-proof pricing strategy in the dynamic housing market.